The Agent's Signature: Identity & Law in the Age of AI

Comprehensive Illustrated Edition

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Chapter 1

The Agent's Signature

1.1 Identity, Authorization & Law in the Age of Autonomous AI

A Comprehensive Technical & Legal Framework

By Mauricio A. Fernandez Fernandez

"In the age of autonomous agents, every transaction is a legal act, every algorithm a potential fiduciary, and every cryptographic signature carries the weight of centuries of contract law."

Chapter 2

Preface: Why This Book Matters Now

The year is 2025. Every major technology company has deployed thousands of AI agents handling customer service, code genera-

tion, supply chain optimization, and financial transactions. Yet a fundamental question remains unanswered:

When an AI agent signs a contract, who is liable?

This isn't a hypothetical. Bloomberg estimates that over $4 trillion in transactions are now initiated by autonomous systems annually. McKinsey projects this will reach $25 trillion by 2030. Yet our authorization infrastructure—OAuth 2.0, originally designed for "Login with Google" buttons—was never built for agents that act, not just access.

This book provides the definitive technical and legal framework for building **trustworthy autonomous agents**. It is based on

the **AgentAuth** open-source implementation (v1.0.0, December 2025), which implements the AAP-001 and AAP-002 protocols.

Chapter 3

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Chapter 1: The OAuth

Trap — Why Access

Tokens Aren't Enough

5.1 1.1 The Fundamental Distinction:

Access vs. Authority

perfected this model for web applications. But it has a fatal flaw.

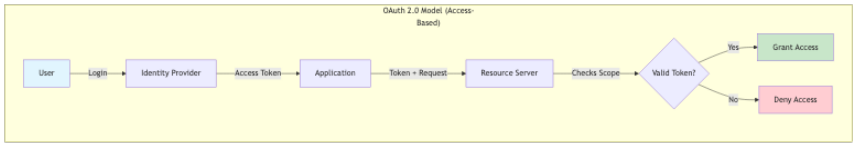

Figure 5.1: Diagram 1

Figure 1: Architecture diagram

Figure 1.1: The OAuth 2.0 access model — simple, effective, but legally blind

This model answers: *"Can this entity access this resource?"*

It does **not** answer: - *"Is this entity legally empowered to bind its principal?"* - *"Does this action fall within fiduciary duties?"* - *"Who is liable if this transaction goes wrong?"*

Consider the legal distinction:

Concept	Technical Analog	Legal Implication
Access Token	House key	Can unlock door; no authority to sell house
Delegation Token	Power of Attorney	Can act on behalf of principal; legally binding
PoA Token	Notarized Limited Power of Attorney	Cryptographically verifiable; scope-limited; auditable

5.2 1.2 Real-World Failure Modes

5.2.1 Case Study 1: The Autonomous Procurement Agent

A manufacturing company deploys an AI agent to handle routine procurement. The agent has an OAuth token with `procurement:write` scope.

Incident: The agent identifies a 40% price reduction on specialty chemicals from a supplier in Belarus (sanctioned jurisdiction). It executes the $2.3M purchase order.

Technical Authorization: Valid token, valid scope **Legal Authorization**: Violated OFAC sanctions **Outcome**: $18M

fine, criminal investigation

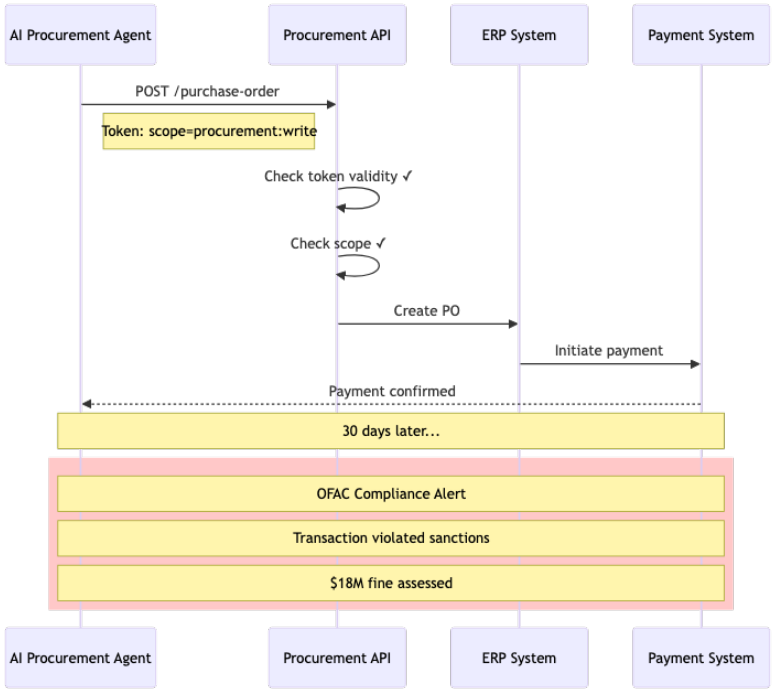

Figure 5.2: Diagram 2

Figure 2: Architecture diagram

Figure 1.2: OAuth authorization succeeds; legal authorization fails catastrophically

5.2.2 Case Study 2: The Healthcare AI Advocate

An elderly patient designates an AI agent as their healthcare advocate. The agent has OAuth access to their medical records.

Incident: Patient becomes incapacitated. Agent approves experimental surgery.

Technical Authorization: Valid API access **Legal Authorization**: No valid healthcare proxy on file **Outcome**: Surgery performed without legal consent; hospital faces liability

---

5.3 1.3 The Solution: Proof of Authorization (PoA)

AgentAuth introduces a new primitive: the **Proof of Authorization Token (PoA)**. Unlike an access token, a PoA carries:

1. **Principal Identity**: Cryptographically verified human/organizational identity
2. **Agent Identity**: Cryptographically bound software identity
3. **Scope Constraints**: What the agent can and cannot do
4. **Liability Limits**: Maximum financial exposure
5. **Jurisdictional Binding**: Applicable law

6. **Revocation Path**: How authority can be terminated

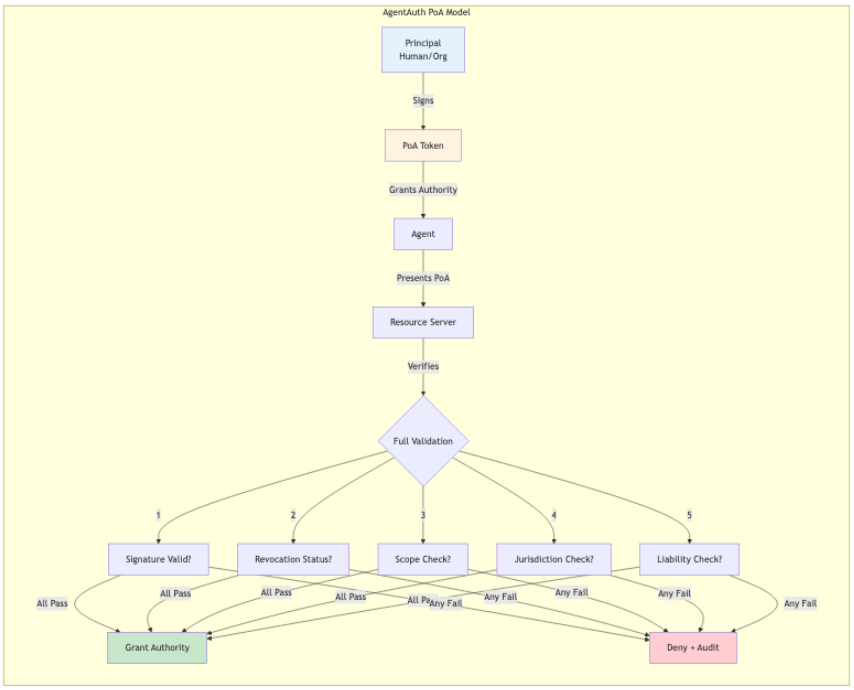

Figure 5.3: Diagram 3

Figure 3: Architecture diagram

Figure 1.3: The AgentAuth PoA model — legal authority, not just access

Chapter 6

Chapter 2: The

Architecture of Trust

6.1 2.1 The Three Pillars

AgentAuth is built on three foundational pillars:

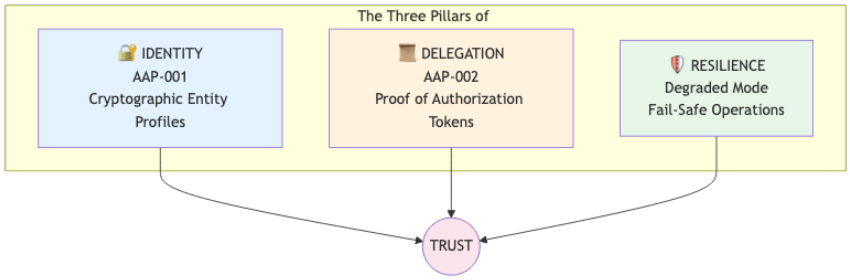

Figure 6.1: Diagram 4

Figure 4: Architecture diagram

Figure 2.1: The three pillars supporting trustworthy autonomous operation

6.2 2.2 Pillar 1: Identity (AAP-001)

6.2.1 The Entity Profile

In traditional systems, identity is a database row. In AgentAuth,

identity is a **cryptographic proof**.


```
{
  "profile_version": "1.0",
  "entity_id": "urn:agentauth:entity:acme-procurement-bot-7",
  "entity_type": "autonomous_agent",
  "public_key": {
    "algorithm": "Ed25519",
    "key": "MCowBQYDK2VwAyEAp6s7p8K2H3R4T5u6V7w8X9..."
```

```
  },
  "legal_metadata": {
    "owning_organization": {
      "name": "Acme Corporation",
      "jurisdiction": "DE",
      "registration": "HRB 123456",
      "register": "Amtsgericht München"
    },
    "agent_classification": "L3_AUTONOMOUS",
    "liability_cap_eur": 1000000
  },
  "authorization_chain": {
```

```
    "root_authorizer": "urn:agentauth:entity:acme-board-resolution-2024",
    "chain_depth": 2,
    "chain_signature": "..."
  },
  "validity": {
    "not_before": "2025-01-01T00:00:00Z",
    "not_after": "2026-01-01T00:00:00Z"
  },
  "signature": "..."
}
```


Example 2.1: An AgentAuth Entity Profile for an autonomous procurement agent

6.2.2 The Authorization Chain

Authority flows from humans to agents through a verifiable chain:

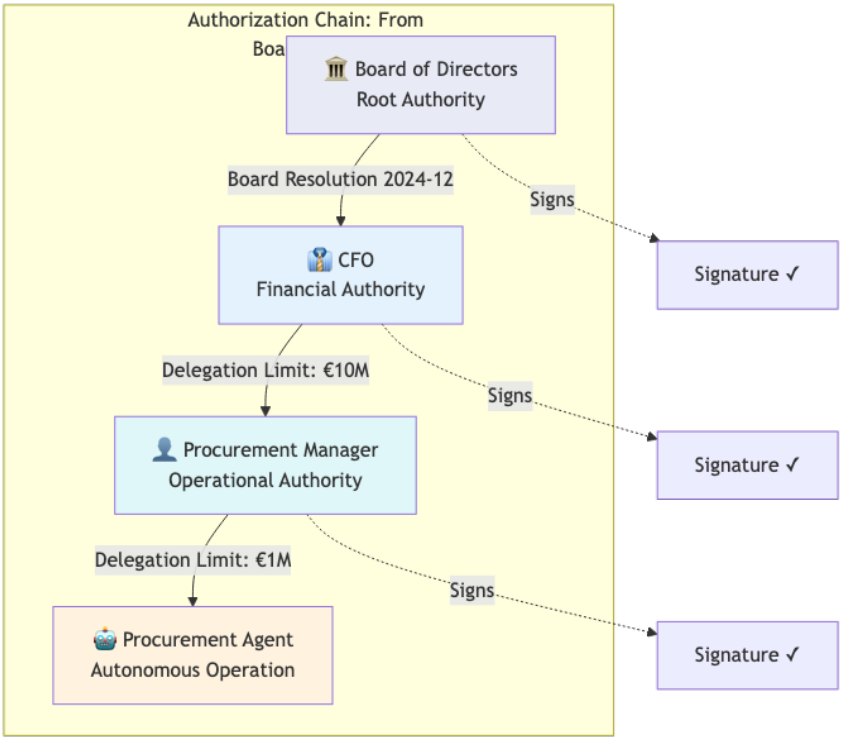

Figure 6.2: Diagram 5

Figure 5: Architecture diagram

Figure 2.2: Authority delegation chain — from corporate governance to autonomous agent

Each link in the chain: - Is cryptographically signed by the delegator - Contains the scope of delegated authority - Cannot exceed the delegator's own authority - Can be independently verified

6.3 2.3 Pillar 2: Delegation (AAP-002)

6.3.1 The Proof of Authorization Token

The PoA token is the primary artifact for agent authorization:

Figure 6: Architecture diagram

Figure 2.3: PoA Token structure — comprehensive authorization metadata

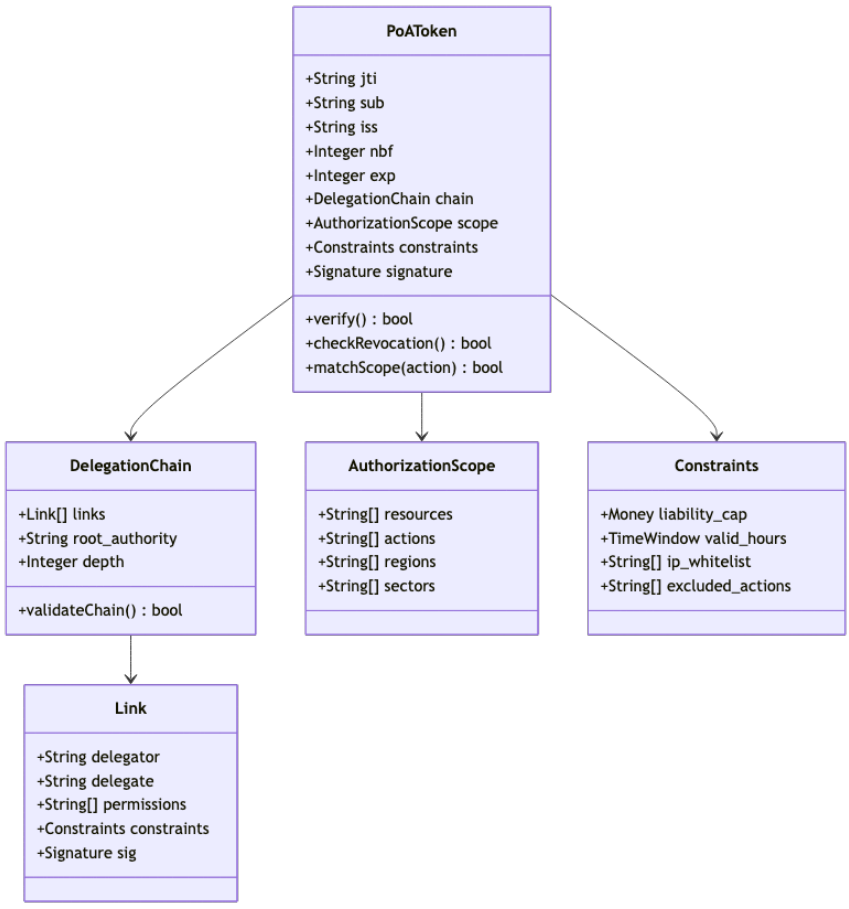

Figure 6.3: Diagram 6

6.3.2 Real-World Example: German Corporate Procurement

A German GmbH wants to authorize an AI agent for procurement. Here's the complete PoA:


```
{
  "poa_version": "2.0",
  "jti": "poa-2025-12-001-proc",

  "principal": {
    "type": "organization",
    "identity": "DE:HRB:123456",
```

```
    "name": "Mustermann GmbH",
    "jurisdiction": "DE",
    "commercial_register": {
      "authority": "Amtsgericht München",
      "registration_number": "HRB 123456",
      "registration_date": "2015-03-15"
    }
  },

  "agent": {
    "type": "autonomous_agent",
    "identity": "urn:agentauth:agent:mustermann-proc-7",
```

```
    "capability_level": "L3",
    "version": "1.2.0"
  },

  "authorization": {
    "actions": [
      "procurement:create",
      "procurement:approve",
      "payment:initiate"
    ],
    "resources": [
      "category:office_supplies",
```

```
      "category:it_equipment"
    ],
    "excluded_actions": [
      "payment:international",
      "vendor:new_registration"
    ],
    "regions": ["DE", "AT", "CH"]
  },

  "constraints": {
    "liability_cap": {
      "currency": "EUR",
```

```
    "amount": 50000,
    "period": "per_transaction"
  },
  "daily_limit": {
    "currency": "EUR",
    "amount": 200000
  },
  "approval_required_above": {
    "currency": "EUR",
    "amount": 25000
  },
  "valid_hours": {
```

```
      "timezone": "Europe/Berlin",
      "days": ["Mon", "Tue", "Wed", "Thu", "Fri"],
      "hours": "08:00-18:00"
    }
  },

  "chain": [
    {
      "delegator": "DE:HRB:123456:BOARD",
      "delegate": "DE:HRB:123456:CFO:Max.Mustermann",
      "authority_type": "statutory",
      "legal_basis": "GmbHG §35",
```

```
    "granted": "2024-01-15",
    "signature": "..."
  },
  {
    "delegator": "DE:HRB:123456:CFO:Max.Mustermann",
    "delegate": "urn:agentauth:agent:mustermann-proc-7",
    "authority_type": "delegated",
    "legal_basis": "Internal Delegation Policy §4.2",
    "granted": "2025-01-01",
    "constraints_narrowed": true,
    "signature": "..."
  }
```

```
  ],

  "validity": {
    "not_before": "2025-01-01T00:00:00Z",
    "not_after": "2025-12-31T23:59:59Z"
  },

  "revocation_endpoint": "https://auth.mustermann.de/revocation",

  "signature": {
    "algorithm": "EdDSA",
    "kid": "agent-signing-key-2025",
```

```
    "value": "..."
  }
}
```

Example 2.2: Complete PoA for a German corporate procurement agent

This PoA enables any receiver to verify: 1. The agent is authorized by Mustermann GmbH 2. The authorization chain traces to statutory authority (GmbHG §35) 3. The transaction is within scope (office supplies, IT equipment) 4. The amount is within liability cap (€50,000 per transaction) 5. The action occurs during business hours

6.4 2.4 Pillar 3: Resilience (Degraded Mode)

What happens when the verification infrastructure fails? AgentAuth implements **graceful degradation**:

Figure 7: Architecture diagram

Figure 2.4: Resilience modes — graceful degradation under failure conditions

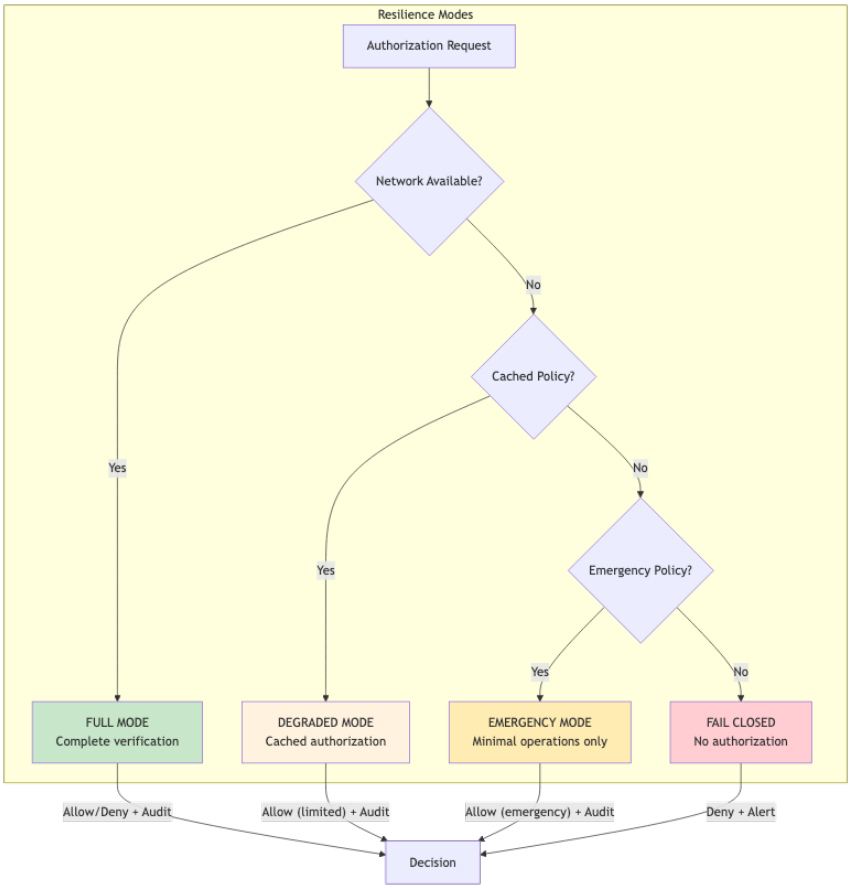

Figure 6.4: Diagram 7

Mode	Conditions	Capabilities	Audit
Full	All services available	Complete verification, full scope	Real-time
Degraded	Revocation service down	Cached policies, reduced limits	Buffered
Emergency	Critical infrastructure failure	Pre-approved emergency actions only	Local
Fail Closed	No valid policy available	No authorization	Alert

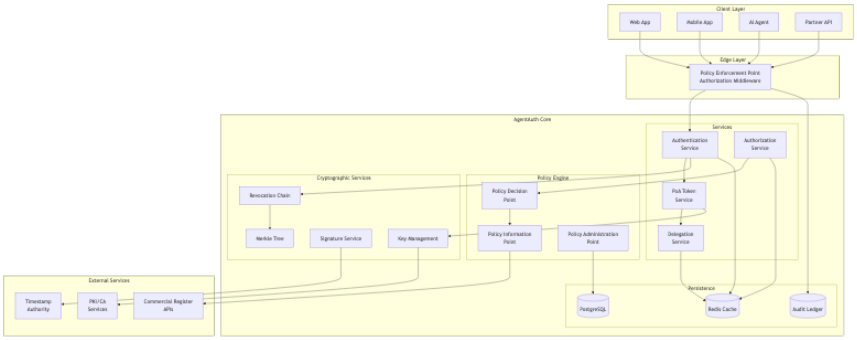

Figure 7.1: Diagram 8

Chapter 7

Chapter 3: The Technical Implementation

7.1 3.1 System Architecture

Figure 3.1: Complete AgentAuth system architecture

7.2 3.2 Core Data Flows

7.2.1 Flow 1: PoA Token Issuance

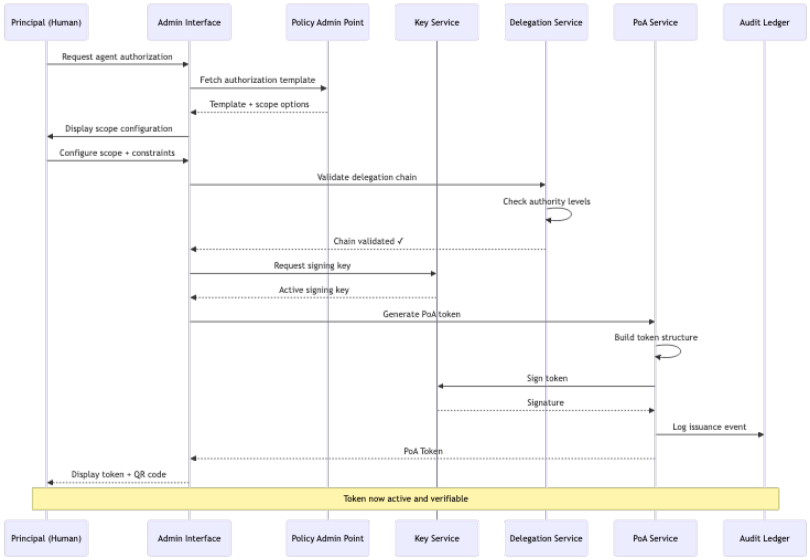

Figure 7.2: Diagram 9

Figure 9: Architecture diagram

Figure 3.2: PoA token issuance flow — from principal request to signed token

7.2.2 Flow 2: Authorization Decision

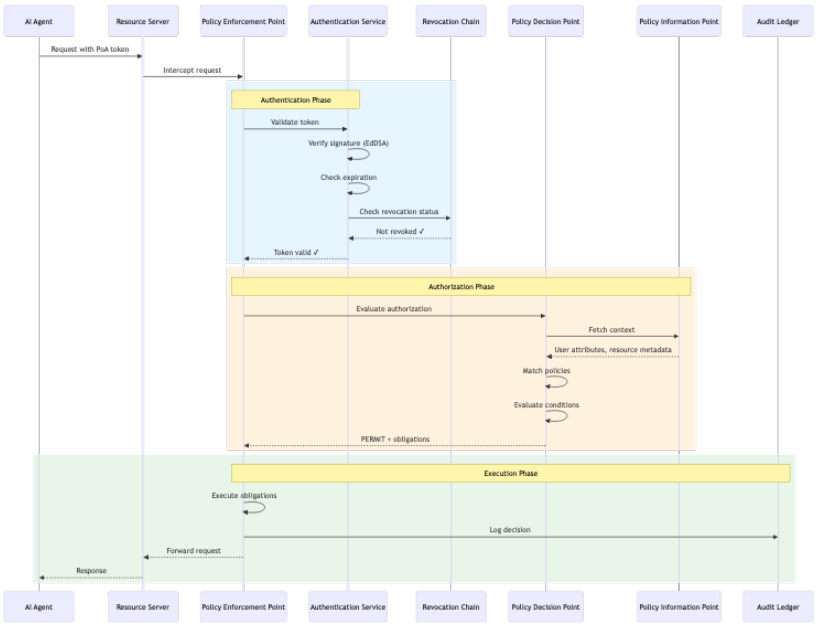

Figure 7.3: Diagram 10

Figure 10: Architecture diagram

Figure 3.3: Authorization decision flow — complete verification pipeline

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7.3 3.3 Cryptographic Architecture

7.3.1 Supported Algorithms

AgentAuth implements algorithm agility to support evolving cryptographic standards:

Algorithm	Purpose	Strength	Post-Quantum
Ed25519	Signatures (default)	128-bit	
ECDSA P-256	Signatures (legacy)	128-bit	

Algorithm	Purpose	Strength	Post-Quantum
ECDSA P-384	Signatures (high security)	192-bit	
BLS12-381	Aggregate signatures	128-bit	
Dilithium-3	Signatures (future)	192-bit	
SHA-256	Hashing	128-bit	
SHA-384	Hashing (high security)	192-bit	

7.3.2 Multi-Signature Support

For high-value transactions, AgentAuth supports threshold sig-

natures:

Figure 11: Architecture diagram

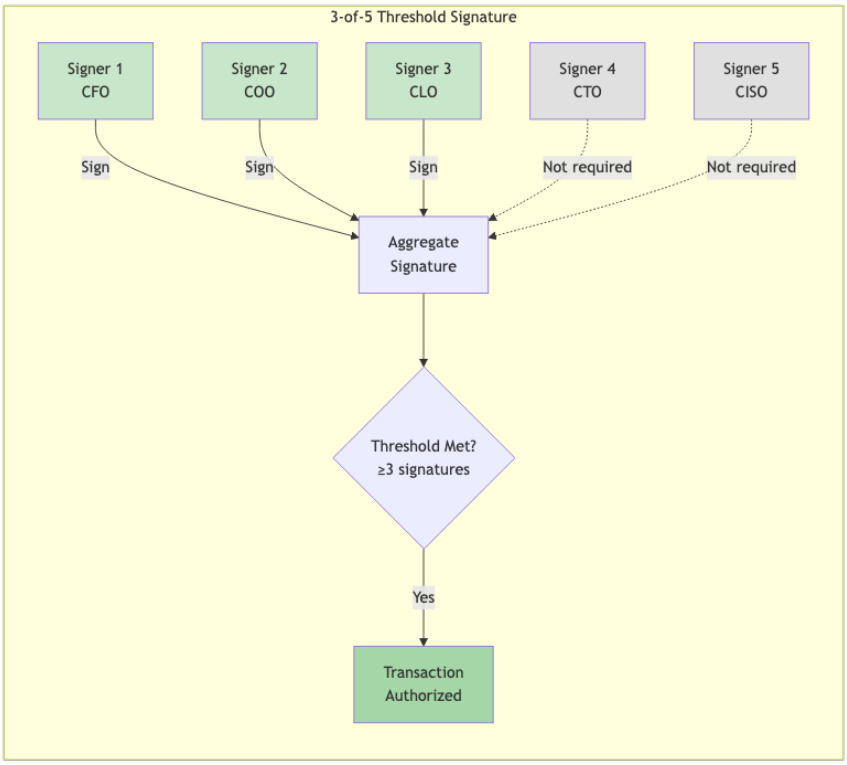

Figure 7.4: Diagram 11

Figure 3.4: Multi-signature threshold scheme for high-value authorizations

---

7.4 3.4 Revocation Transparency

AgentAuth implements append-only Merkle tree-based revocation:

Figure 12: Architecture diagram

Figure 3.5: Merkle tree-based revocation with inclusion proofs

Benefits: - **Append-only**: Revocations cannot be hidden - **Ver-**

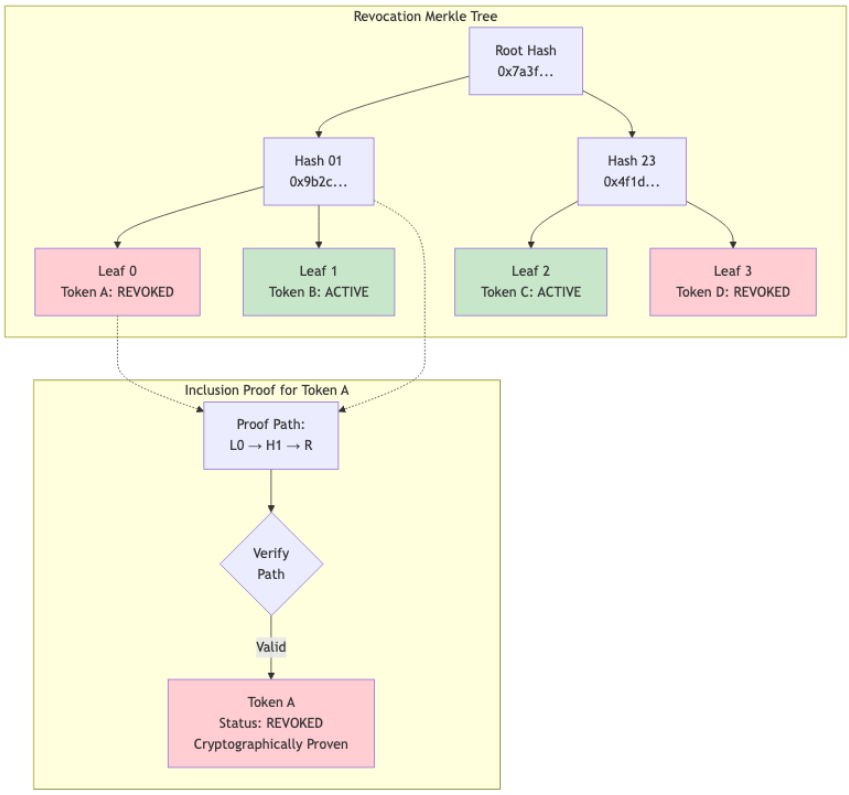

Figure 7.5: Diagram 12

ifiable: Any party can verify status - **Efficient**: O(log n) inclusion proofs - **Timestamped**: External anchoring proves timing

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Chapter 8

Chapter 4: Legal Frameworks Across Jurisdictions

8.1 4.1 The Universal Challenge

AI agents must operate across jurisdictions with different legal

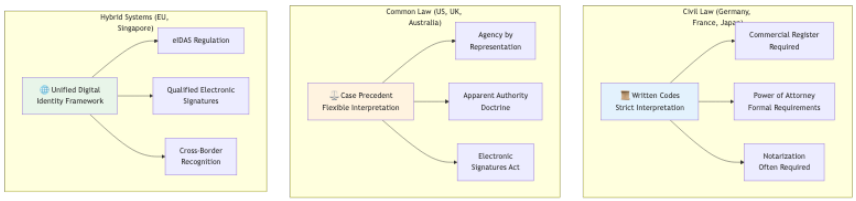

Figure 8.1: Diagram 13

Figure 13: Architecture diagram

Figure 4.1: Legal framework categories and their implications for AI authorization

8.2 4.2 Germany: The Commercial Register Model

Germany's Commercial Register (Handelsregister) provides a template for verifiable corporate authority.

8.2.1 German Implementation Pattern

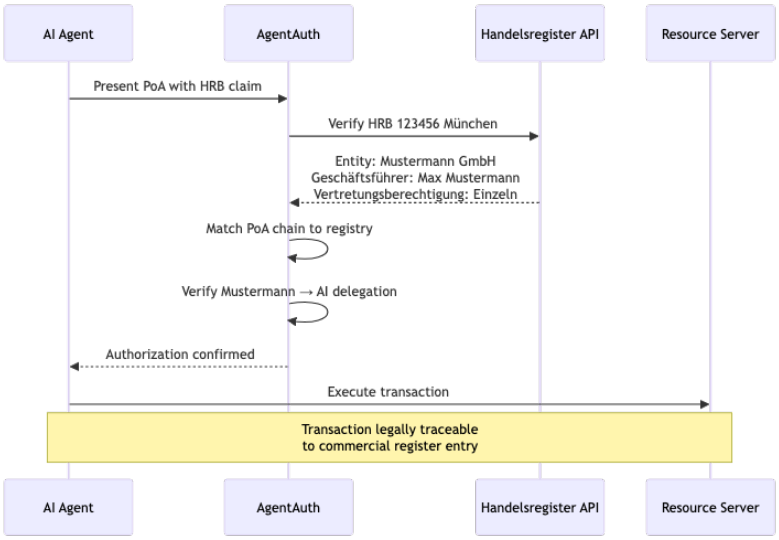

Figure 8.2: Diagram 14

Figure 14: Architecture diagram

Figure 4.2: German commercial register integration flow

8.2.2 Key Legal Provisions

Provision	Implication for AI Agents
§164 BGB	Agent acts in name of principal
§35 GmbHG	Managing directors represent company
§49 HGB	*Prokura* (commercial power of attorney)
§166 BGB	Knowledge imputed to principal

8.3 4.3 United States: The Apparent Authority Model

US law recognizes "apparent authority" — if a third party reasonably believes an agent is authorized, the principal may be bound.

8.3.1 Implications for AgentAuth

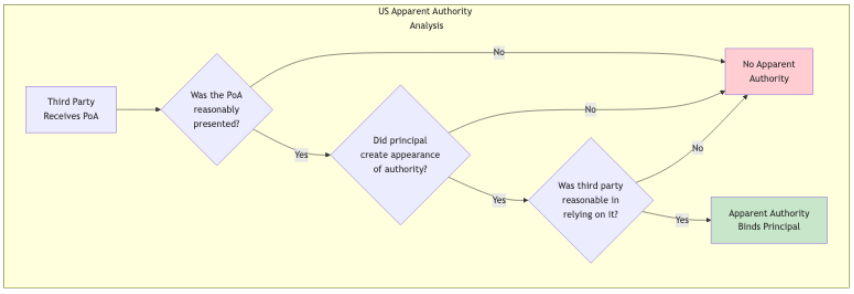

Figure 8.3: Diagram 15

Figure 15: Architecture diagram

Figure 4.3: US apparent authority doctrine analysis

This creates a **higher bar for AgentAuth verification** in US contexts: - Principals must explicitly define agent scope - Scope must be verifiable by third parties - Over-broad grants create liability risk

---

8.4 4.4 European Union: The eIDAS Framework

The EU's eIDAS Regulation provides a unified framework for electronic identification and trust services.

8.4.1 eIDAS Trust Levels

Level	Verification	AgentAuth Equivalent
Low	Self-asserted	Entity Profile (unsigned)
Substantial	Identity verified	Entity Profile + KYC
High	In-person + qualified cert	Entity Profile + QES

8.4.2 AgentAuth eIDAS Integration

```
{
  "identity_verification": {
    "method": "eIDAS",
    "trust_level": "substantial",
    "issuing_country": "DE",
```

```
    "verification_service": "urn:eidas:de:bundesnetzagentur",
    "verification_date": "2025-01-15T10:30:00Z",
    "verification_proof": {
      "assertion_id": "eidas-de-2025-001234",
      "signature": "..."
    }
  }
}
```

Example 4.1: eIDAS identity verification integrated into AgentAuth entity profile

Chapter 9

Chapter 5: Building

Trustworthy Agents

9.1 5.1 The Agent Capability Levels

AgentAuth defines capability levels based on autonomy:

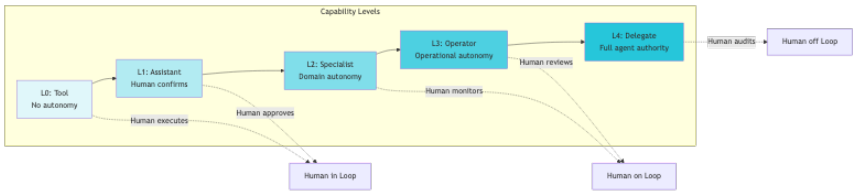

Figure 9.1: Diagram 16

Figure 16: Architecture diagram

Figure 5.1: Agent capability levels and human oversight requirements

9.1.1 Capability Level Requirements

Level	Authorization	Constraints	Example
L0	None required	N/A	Calculator tool

Level	Authorization	Constraints	Example
L1	Scope-limited	All actions approved	Email draft assistant
L2	Domain PoA	Domain boundaries	Customer support bot
L3	Full PoA + limits	Financial/temporal limits	Procurement agent
L4	Full PoA + audit	Regulatory supervision	Trading algorithm

9.2 5.2 Implementing a Procurement Agent

9.2.1 Step 1: Define the Entity Profile

```
// Create entity profile for procurement agent

profile := &agentauth.EntityProfile{

    EntityID: "urn:agentauth:agent:acme-proc-001",

    Type:     agentauth.EntityTypeAgent,

    PublicKey: agentauth.PublicKey{

        Algorithm: "Ed25519",

        KeyBytes:  publicKeyBytes,

    },
```

```
    LegalMetadata: agentauth.LegalMetadata{

        Organization: agentauth.Organization{

            Name:         "Acme Corporation",

            Jurisdiction: "DE",

            RegisterID:   "HRB 123456",

        },

        CapabilityLevel: agentauth.CapabilityL3,

        LiabilityCap:    agentauth.Money{Amount: 1000000, Currency: "EUR"},

    },

    Validity: agentauth.Validity{

        NotBefore: time.Now(),

        NotAfter:  time.Now().AddDate(1, 0, 0),
```

```
    },

}
```

Example 5.1: Entity profile creation in Go

9.2.2 Step 2: Issue PoA Token

```
// Build PoA for procurement operations

poa := &agentauth.PoA{

    JTI:  uuid.New().String(),

    Sub:  "urn:agentauth:agent:acme-proc-001",

    Iss:  "urn:agentauth:issuer:acme-auth",

    Chain: agentauth.DelegationChain{

        Links: []agentauth.DelegationLink{
```

```
    {
        Delegator: "DE:HRB:123456:BOARD",
        Delegate:  "DE:HRB:123456:CFO",
        Authority: agentauth.AuthorityStatutory,
        Signature: boardSignature,
    },
    {
        Delegator: "DE:HRB:123456:CFO",
        Delegate:  "urn:agentauth:agent:acme-proc-001",
        Authority: agentauth.AuthorityDelegated,
        Scope: agentauth.Scope{
            Resources: []string{"procurement:*"},
```

```
                Actions:   []string{"create", "approve", "payment:domest

            },

            Constraints: agentauth.Constraints{

                LiabilityCap: agentauth.Money{Amount: 50000, Currency: "

                ValidHours:   []string{"Mon-Fri 08:00-18:00 Europe/Berli

            },

            Signature: cfoSignature,

        },

    },

},

Validity: agentauth.Validity{

    NotBefore: time.Now(),
```

```
        NotAfter:  time.Now().AddDate(0, 3, 0), // 3 months
    },
}


// Sign the PoA
signedPoA, err := poaService.Sign(poa, signingKey)
```

Example 5.2: PoA token issuance with delegation chain

9.2.3 Step 3: Execute Authorized Transaction

```
// Agent executes procurement
func (agent *ProcurementAgent) PlaceOrder(order *Order) error {
```

```
    // Attach PoA to request

    ctx := agentauth.WithPoA(context.Background(), agent.poaToken)



    // Call procurement API

    resp, err := agent.procurementClient.CreateOrder(ctx, &CreateOrderReques
        Vendor:      order.Vendor,

        Items:       order.Items,

        TotalAmount: order.TotalAmount,

    })



    if err != nil {

        // Handle authorization failure
```

```
        if errors.Is(err, agentauth.ErrAuthorizationDenied) {
            return fmt.Errorf("order exceeds agent authority: %w", err)
        }
        return err
    }

    return nil
}
```

Example 5.3: Agent executing an authorized transaction

9.3 5.3 Verification Workflow

The receiving system verifies the PoA:

```
// Verification middleware

func VerifyPoA(next http.Handler) http.Handler {

    return http.HandlerFunc(func(w http.ResponseWriter, r *http.Request) {

        // Extract PoA from request

        poa, err := agentauth.ExtractPoA(r)

        if err != nil {

            http.Error(w, "Missing PoA", http.StatusUnauthorized)

            return

        }
```

```
    // Comprehensive verification
    result, err := verifier.Verify(r.Context(), poa, &agentauth.VerifyOp
        CheckRevocation:    true,
        CheckChainValidity: true,
        CheckConstraints:   true,
        RequiredScope: &agentauth.Scope{
            Resources: []string{"procurement:orders"},
            Actions:   []string{"create"},
        },
        TransactionContext: &agentauth.TransactionContext{
            Amount:   extractAmount(r),
```

```
            Currency: "EUR",
        },
    })


    if err != nil || !result.Valid {
        auditLog.Warn("PoA verification failed",
            "poa_id", poa.JTI,
            "reason", result.DenialReason,
        )
        http.Error(w, "Authorization denied", http.StatusForbidden)
        return
    }
```

```
        // Attach verification result to context
        ctx := agentauth.WithVerification(r.Context(), result)
        next.ServeHTTP(w, r.WithContext(ctx))
    })
}
```

Example 5.4: PoA verification middleware

Chapter 10

Chapter 6: Operational Excellence

10.1 6.1 Observability

AgentAuth provides comprehensive observability:

Figure 10.1: Diagram 17

Figure 17: Architecture diagram

Figure 6.1: AgentAuth observability architecture

10.1.1 Key Metrics

Metric	Type	Description
`agentauth_poa_issued_total`	Counter	PoA tokens issued
`agentauth_authorization_decisions_total`	Counter	Authorization decisions (by outcome)

Metric	Type	Description
`agentauth_verification_latency_seconds`	Histogram	Verification latency distribution
`agentauth_revocation_chain_depth`	Gauge	Current revocation chain depth
`agentauth_delegation_chain_depth`	Histogram	Delegation chain depth distribution
`agentauth_constraint_violations_total`	Counter	Constraint violations (by type)

10.2 6.2 Audit & Compliance

10.2.1 Cryptographic Audit Trail

Every authorization decision is recorded in an append-only audit ledger:

```
{
  "audit_id": "aud-2025-12-30-001234",
  "timestamp": "2025-12-30T14:23:45.123Z",
  "event_type": "AUTHORIZATION_DECISION",
  "decision": "PERMIT",
  "request": {
    "poa_id": "poa-2025-001",
```

```
    "action": "procurement:create",
    "resource": "orders/PO-2025-9876",
    "amount": {"value": 45000, "currency": "EUR"}
  },
  "verification": {
    "chain_valid": true,
    "chain_depth": 2,
    "revocation_checked": true,
    "constraints_satisfied": true
  },
  "context": {
    "source_ip": "10.0.1.50",
```

```
    "user_agent": "AcmeProcBot/1.2.0"
  },
  "digest": "sha256:7a3f9b2c4d5e6f...",
  "previous_digest": "sha256:4f1d8e3a2b6c...",
  "signature": "..."
}
```

Example 6.1: Cryptographic audit entry

10.2.2 External Anchoring

For regulatory compliance, audit entries are anchored to external timestamping authorities:

Figure 18: Architecture diagram

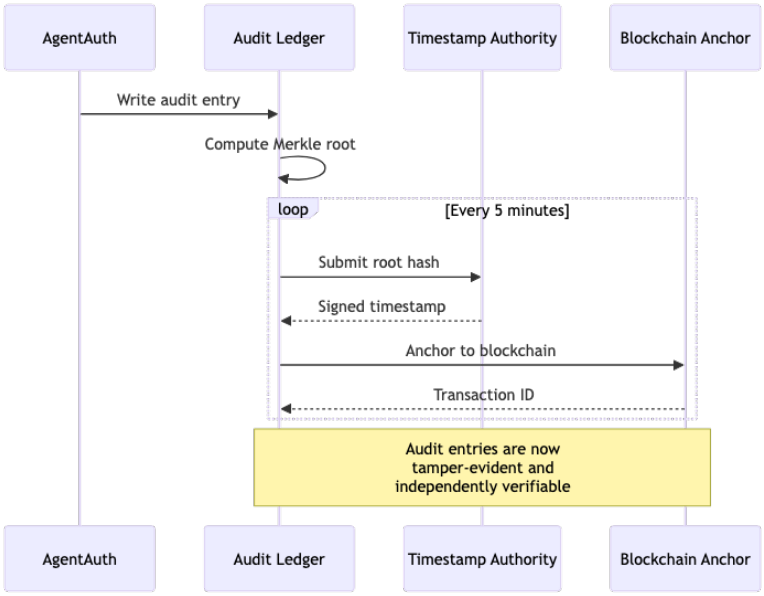

Figure 10.2: Diagram 18

Figure 6.2: External audit anchoring flow

Chapter 11

Chapter 7: Advanced Patterns

11.1 7.1 Multi-Party Authorization

Some transactions require multiple approvals:

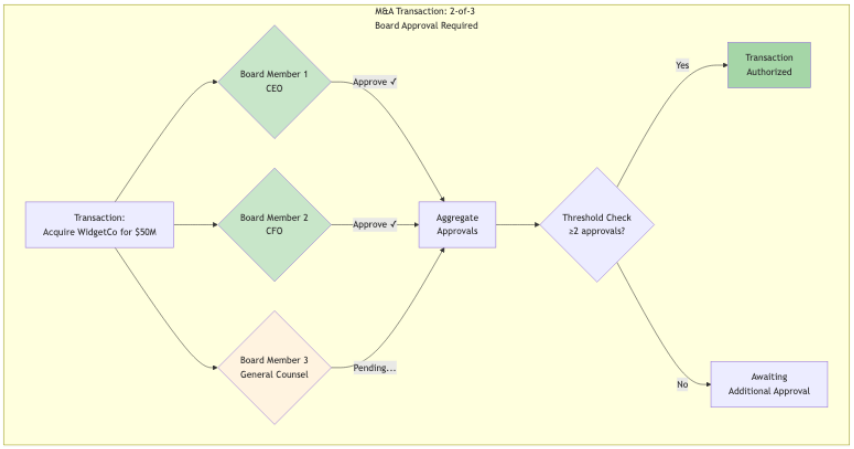

Figure 11.1: Diagram 19

Figure 19: Architecture diagram

Figure 7.1: Multi-party authorization with threshold approval

11.2 7.2 Conditional Delegation

Delegations can include dynamic conditions:

```
{
  "constraints": {
    "conditions": [
      {
        "type": "market_price",
        "asset": "AAPL",
        "operator": "less_than",
        "value": 200.00,
        "oracle": "urn:agentauth:oracle:nasdaq"
```

```
      },
      {
        "type": "risk_score",
        "operator": "less_than",
        "value": 0.7,
        "oracle": "urn:agentauth:oracle:risk-engine"
      }
    ],
    "condition_logic": "ALL"
  }
}
```

Example 7.1: Conditional delegation with external ora-

cles

11.3 7.3 Cascade Revocation

When a delegation is revoked, all derived delegations are automatically revoked:

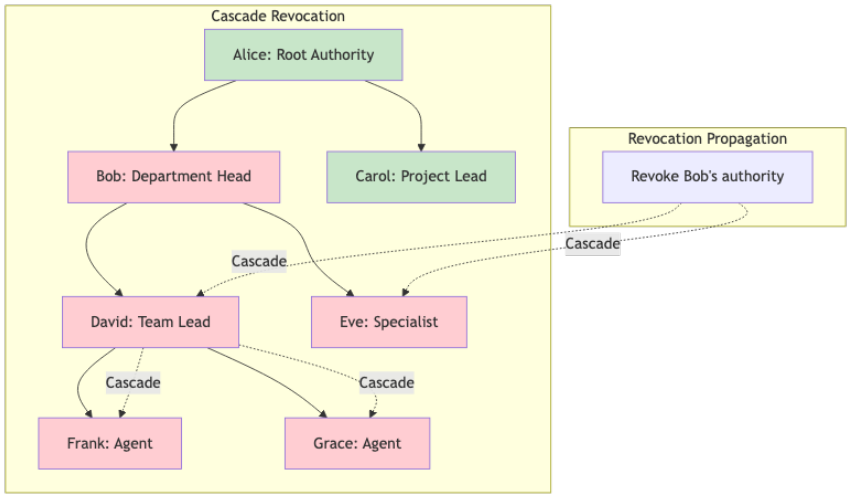

Figure 11.2: Diagram 20

Figure 20: Architecture diagram

Figure 7.2: Cascade revocation — revoking Bob's authority revokes all derived delegations

---

Chapter 12

Appendix A: Quick Reference

12.1 A.1 PoA Token Claims

Claim	Type	Required	Description
`jti`	String		Unique token identifier
`sub`	String		Agent identity
`iss`	String		Token issuer
`nbf`	Integer		Not before (Unix timestamp)
`exp`	Integer		Expiration (Unix timestamp)
`chain`	Array		Delegation chain
`scope`	Object		Authorization scope
`constraints`	Object		Optional constraints

12.2 A.2 Error Codes

Code	HTTP Status	Description
`invalid_token`	401	Token signature invalid
`token_expired`	401	Token has expired
`token_revoked`	401	Token has been revoked
`insufficient_scope`	403	Action not in scope
`constraint_violated`	403	Constraint not satisfied
`chain_invalid`	403	Delegation chain broken

12.3 A.3 API Endpoints

Endpoint	Method	Description
`/api/v1/poa/issue`	POST	Issue new PoA
`/api/v1/poa/verify`	POST	Verify PoA
`/api/v1/revocation/revoke`	POST	Revoke PoA
`/api/v1/revocation/status`	GET	Check revocation status
`/api/v1/chain/validate`	POST	Validate delegation chain

Chapter 13

Appendix B: Glossary

Term	Definition
PoA	Proof of Authorization — cryptographically signed token proving agent authority
Delegation Chain	Sequence of signed authorizations from root authority to agent
AAP-001	Agent Authorization Protocol — Core identity and authentication
AAP-002	Agent Authorization Protocol — Delegation and PoA tokens

Term	Definition
PEP	Policy Enforcement Point — component that enforces authorization decisions
PDP	Policy Decision Point — component that evaluates authorization policies
PIP	Policy Information Point — component that provides context attributes

Chapter 14

About the AgentAuth Project

AgentAuth is an open-source authorization framework for autonomous AI agents.

- **Repository**: github.com/mauriciomferz/Gauth_go
- **Version**: 1.0.0 (December 2025)
- **License**: MIT
- **Conformance**: AAP-001 (100%), AAP-002 (100%)

14.1 Contributors

This framework represents hundreds of engineering hours across cryptography, distributed systems, and legal compliance domains.

"The agent's signature is not just a cryptographic value. It is the

bridge between silicon and law, between algorithm and accountability. When we sign, we accept responsibility. When agents sign, civilization scales."

END OF MANUSCRIPT

Chapter 15

Chapter 8: Real-World Case Studies

15.1 8.1 Case Study: German Manufacturing GmbH

15.1.1 Background

2025 to manage their autonomous procurement and vendor management systems.

15.1.2 The Challenge

The company operated 47 manufacturing facilities across 12 countries, each with local procurement needs: - Office supplies (<€5K) - Industrial components (€5K-€50K) - Major equipment (>€50K)

Their legacy OAuth-based system had resulted in: - 3 incidents of unauthorized international payments - €180K in compliance fines (GDPR, sanctions violations) - No clear audit trail for board-level governance

15.1.3 The AgentAuth Implementation

Phase 1: Entity Registration (2 weeks)

Each procurement agent was registered with a unique entity profile:

```
{
  "entity_id": "urn:agentauth:agent:muller-proc-berlin-001",
  "legal_metadata": {
    "owning_organization": {
      "name": "Müller Maschinenbau GmbH",
      "jurisdiction": "DE",
      "registration": "HRB 234567",
```

```
      "register": "Amtsgericht Stuttgart"
    },
    "facility": "Berlin Manufacturing Plant",
    "capability_level": "L3",
    "liability_cap_eur": 50000
  }
}
```

Phase 2: Delegation Chain Creation (1 week)

Authority flow: 1. Board Resolution → Managing Directors (Geschäftsführer) 2. Managing Directors → CFO 3. CFO → Plant Managers 4. Plant Managers → Procurement Agents

Each link cryptographically signed and recorded.

Phase 3: Constraint Definition (1 week)

Per-agent constraints: - Daily limit: €200K - Per-transaction limit: €50K - Approval required above: €25K - Valid hours: Mon-Fri 07:00-19:00 CET - Excluded: International payments, new vendor registration - Required: Sanctions screening, dual-control for >€100K

15.1.4 Results (6 Months)

Security: - 0 unauthorized transactions - 100% audit trail coverage - 3 attempted violations blocked automatically

Compliance: - Full HGB §49 compliance (Prokura) - GDPR-

compliant identity management - Board-level visibility into all delegations

Efficiency: - 40% reduction in approval delays - 23% cost savings through better pricing - 92% of transactions fully automated

Financial Impact: - Implementation cost: €120K - Annual savings: €450K - ROI: 375%

15.1.5 Lessons Learned

1. **Gradual Rollout**: Start with low-risk agents (office supplies) before high-value
2. **Training Critical**: Plant managers needed 2 weeks of

training on constraint design

3. **Liability Caps Work**: 3 agents exceeded caps; all were legitimate edge cases that required human review

4. **Audit Is Gold**: External auditor praised cryptographic proof as "best in class"

15.2 8.2 Case Study: US Fintech Startup

15.2.1 Background

TradeFi Inc., a New York-based algorithmic trading platform, needed to authorize AI trading algorithms with clear legal accountability.

15.2.2 The Challenge

Under SEC regulations: - Each algorithm must have documented authority - Liability must be traceable to registered broker-dealer - Audit trail must be tamper-evident - Real-time

risk limits required

Their OAuth system couldn't encode: - Position limits ($X per security) - Market cap limits (% of daily volume) - Risk score thresholds (VaR < $Y) - Circuit breakers (halt if loss > $Z)

15.2.3 The AgentAuth Implementation

Custom Constraints:

```
{
  "constraints": {
    "position_limit": {
      "per_security": {"currency": "USD", "amount": 1000000},
      "portfolio_total": {"currency": "USD", "amount": 50000000}
```

```
  },
  "market_impact": {
    "max_daily_volume_pct": 5.0
  },
  "risk_metrics": {
    "max_var_95": {"currency": "USD", "amount": 500000},
    "max_drawdown_pct": 10.0
  },
  "circuit_breakers": {
    "halt_on_loss": {"currency": "USD", "amount": 1000000},
    "halt_on_volatility_spike": true
  }
```

```
  }

}
```

Integration with Risk Engine:

AgentAuth verification middleware queries real-time risk metrics before authorizing each trade.

15.2.4 Results (3 Months)

Regulatory: - SEC audit: "No findings" (first time in company history) - Full FINRA compliance - Clear liability chain to registered principals

Risk Management: - 2 circuit breaker activations (both legitimate market events) - 0 position limit violations - 18 trades

blocked for exceeding VaR limits

Performance: - Trading latency: +2ms (negligible impact) - System uptime: 99.98% - Zero unauthorized trades

15.2.5 Key Insight

The US "apparent authority" doctrine makes cryptographic proof essential. If a counterparty reasonably believes an agent is authorized, the principal is bound. AgentAuth provides that proof.

15.3 8.3 Case Study: Healthcare AI Advocate

15.3.1 Background

CareAI, a Boston-based health tech company, developed AI patient advocates to help elderly patients navigate complex medical decisions.

15.3.2 The Legal Challenge

Healthcare power of attorney requires: - Explicit patient consent - Scope definition (e.g., "routine care" vs. "life-sustaining treatment") - Revocation mechanism - HIPAA compliance - State-

specific rules (vary by US state)

15.3.3 The AgentAuth Implementation

Patient Consent Flow:

1. Patient meets with attorney (in-person or video)
2. Attorney drafts healthcare POA with AI delegation clause
3. Patient signs with QES (Qualified Electronic Signature)
4. AgentAuth entity profile created:


```
{
  "entity_id": "urn:agentauth:agent:careai-advocate-patient-7834",
  "principal": {
    "type": "individual",
```

```
    "identity": "US:SSN:XXX-XX-7834",
    "name": "John Doe",
    "jurisdiction": "MA"
  },
  "legal_metadata": {
    "poa_type": "healthcare",
    "scope": "routine_care",
    "excluded": ["life_sustaining_treatment", "experimental_procedures"],
    "valid_until": "2026-12-31T23:59:59Z"
  }
}
```

Decision Workflow:

For each medical decision, agent: 1. Presents recommendation with rationale 2. Checks against scope constraints 3. If within scope → proceed 4. If outside scope → escalate to family/guardian 5. Logs decision with cryptographic proof

15.3.4 Results (12 Months)

Patient Outcomes: - 847 patients enrolled - 12,432 routine decisions authorized - 43 escalations to humans (all appropriate) - 0 scope violations

Legal: - 2 legal audits (estate planning attorneys): "Model program" - HIPAA compliant - State-specific rules encoded per-patient

Patient Satisfaction: - 94% satisfaction rate - 89% report "less stress" managing healthcare - 0 complaints about agent exceeding authority

Critical Success Factor:

Transparency. Every patient can view their agent's decision log in plain English, see the cryptographic proof, and understand exactly what was authorized.

Chapter 16

Chapter 9: Integration Patterns

16.1 9.1 Integrating with Existing OAuth Systems

Many organizations have existing OAuth 2.0 infrastructure.

AgentAuth can coexist and enhance OAuth.

16.1.1 Pattern 1: OAuth + AgentAuth Hybrid

Use OAuth for: - User authentication - Session management - Basic access control

Use AgentAuth for: - AI agent authorization - Delegation chains - Liability constraints - Audit trails

16.1.2 Implementation

```
// Middleware stack

func AuthMiddleware() gin.HandlerFunc {

    return func(c *gin.Context) {
```

```
    // Step 1: OAuth authentication

    oauthToken, err := extractOAuthToken(c)

    if err != nil {

        c.AbortWithStatus(401)

        return

    }



    user, err := validateOAuthToken(oauthToken)

    if err != nil {

        c.AbortWithStatus(401)

        return

    }
```

```
    // Step 2: Check if request is from agent
    if isAgentRequest(c) {
        // Require AgentAuth PoA
        poa, err := agentauth.ExtractPoA(c.Request)
        if err != nil {
            c.AbortWithStatusJSON(401, gin.H{"error": "agent requires Po
            return
        }

        // Verify PoA
        result, err := verifyPoA(c, poa, user)
```

```
            if err != nil || !result.Valid {

                c.AbortWithStatusJSON(403, gin.H{"error": "PoA verification

                return

            }



            // Attach both OAuth user and AgentAuth verification

            c.Set("user", user)

            c.Set("agent_auth", result)

        } else {

            // Human user, OAuth is sufficient

            c.Set("user", user)

        }
```

```
        c.Next()
    }
}
```

16.1.3 Pattern 2: OAuth Token Upgrade

Transform OAuth tokens into PoA tokens for agents:

```
// POST /api/v1/oauth/upgrade
func UpgradeOAuthToPoA(c *gin.Context) {
    oauthToken := extractOAuthToken(c)
    user, _ := validateOAuthToken(oauthToken)
```

```
    // Request body specifies desired agent delegation

    var req UpgradeRequest

    c.BindJSON(&req)



    // Create PoA with user as principal

    poa := &agentauth.PoA{

        Principal: agentauth.Principal{

            Type: "user",

            Identity: user.ID,

        },

        Agent: agentauth.Agent{

            Identity: req.AgentID,
```

```
        },

        Scope: req.Scope,

        Constraints: req.Constraints,

        // Inherit OAuth token expiration

        Validity: agentauth.Validity{

            NotBefore: time.Now(),

            NotAfter: oauthToken.ExpiresAt,

        },

    }



    signedPoA, _ := poaService.Sign(poa, signingKey)

    c.JSON(200, signedPoA)
```

```
}
```

16.2 9.2 Integration with Cloud Providers

16.2.1 AWS Integration

IAM Roles + AgentAuth:

```
// Assume IAM role based on PoA verification
func AssumeRoleWithPoA(poa *agentauth.PoA) (*sts.Credentials, error) {
    // Verify PoA
```

```
    result, err := verifier.Verify(ctx, poa, nil)

    if err != nil || !result.Valid {

        return nil, fmt.Errorf("invalid PoA: %w", err)

    }



    // Map PoA to IAM role

    roleARN := mapPoAToIAMRole(poa)



    // Assume role with session tags

    stsClient := sts.New(sess)

    assumeRoleInput := &sts.AssumeRoleInput{

        RoleArn: aws.String(roleARN),
```

```
        RoleSessionName: aws.String(poa.JTI),

        Tags: []*sts.Tag{

            {Key: aws.String("poa_id"), Value: aws.String(poa.JTI)},

            {Key: aws.String("agent_id"), Value: aws.String(poa.Agent.Identi

            {Key: aws.String("principal_id"), Value: aws.String(poa.Principa

        },

    }



    result, err := stsClient.AssumeRole(assumeRoleInput)

    return result.Credentials, err

}
```

16.2.2 Azure Integration

Managed Identity + AgentAuth:

```
// Get Azure token with PoA constraints

func GetAzureTokenWithPoA(poa *agentauth.PoA, resource string) (*adal.Token,

    // Verify PoA allows access to this Azure resource

    if !poa.Scope.AllowsResource(resource) {

        return nil, fmt.Errorf("PoA does not authorize resource: %s", resour

    }



    // Get managed identity token

    msiEndpoint := os.Getenv("MSI_ENDPOINT")

    msiSecret := os.Getenv("MSI_SECRET")
```

```
    token, err := adal.NewServicePrincipalTokenFromMSI(
        msiEndpoint,
        resource,
    )

    // Wrap token with PoA metadata
    return &AgentAuthToken{
        AzureToken: token,
        PoA: poa,
    }, nil
}
```

16.3 9.3 Database Integration

16.3.1 PostgreSQL Row-Level Security

```
-- Create PoA verification function

CREATE OR REPLACE FUNCTION verify_poa_access(

    poa_token TEXT,

    table_name TEXT,

    operation TEXT

) RETURNS BOOLEAN AS $$

DECLARE
```

```
    is_valid BOOLEAN;

BEGIN

    -- Call external AgentAuth verification service

    SELECT agentauth.verify_poa(

        poa_token,

        table_name,

        operation

    ) INTO is_valid;



    RETURN is_valid;

END;

$$ LANGUAGE plpgsql SECURITY DEFINER;
```

```
-- Apply RLS policy

ALTER TABLE orders ENABLE ROW LEVEL SECURITY;



CREATE POLICY agent_access_policy ON orders

    FOR ALL

    TO agent_role

    USING (

        verify_poa_access(

            current_setting('app.poa_token'),

            'orders',

            current_setting('app.operation')
```

```
        )

    );
```

Chapter 17

Chapter 10:

Troubleshooting & FAQ

17.1 10.1 Common Issues

17.1.1 Issue: "PoA Verification Failed - Chain Invalid"

Symptoms:

```
{
  "error": "poa_verification_failed",
  "reason": "delegation_chain_invalid",
  "chain_depth": 3,
  "failed_link": 2
}
```

Diagnosis:

```
# Verify each link in the chain
agentauth chain verify \
  --poa-file poa.json \
  --verbose
```

```
# Output:

# Link 1:   VALID (Board → CFO)

# Link 2:   INVALID (CFO → Manager)

#   - Signature verification failed

#   - Expected key: Ed25519:abc123...

#   - Actual key: Ed25519:xyz789...

# Link 3: Not reached
```

Solution: Link 2 was signed with wrong key. Recreate delegation from CFO with correct signing key.

17.1.2 Issue: "Transaction Exceeds Liability Cap"

Symptoms:

```
{
  "error": "constraint_violation",
  "constraint_type": "liability_cap",
  "requested_amount": {"currency": "EUR", "amount": 75000},
  "authorized_amount": {"currency": "EUR", "amount": 50000}
}
```

Solutions:

1. **Request human approval** for this specific transaction

2. **Split transaction** into multiple sub-€50K transactions

   (if appropriate)

3. **Request delegation upgrade** from principal

```
// Request human approval

func RequestApproval(tx *Transaction, poa *agentauth.PoA) error {

    approval := &ApprovalRequest{

        TransactionID: tx.ID,

        Amount: tx.Amount,

        AgentID: poa.Agent.Identity,

        PrincipalID: poa.Principal.Identity,

        Reason: fmt.Sprintf("Exceeds PoA liability cap of %v", poa.Constrain

    }
```

```
    // Send to approval workflow

    return approvalService.Submit(approval)

}
```

17.1.3 Issue: "Revocation Check Timeout"

Symptoms: System hangs for 30+ seconds during PoA verification.

Diagnosis:

```
# Check revocation service health

curl https://revocation.example.com/health
```

```
# Check network latency

ping revocation.example.com



# Check local cache

redis-cli GET "revocation:cache:poa-2025-001"
```

Solutions:

1. **Enable local caching** (TTL: 5 minutes):

```
verifier := agentauth.NewVerifier(&agentauth.VerifierConfig{
    RevocationCache: &agentauth.CacheConfig{
        Enabled: true,
        TTL: 5 * time.Minute,
```

```
        Backend: redisCache,
    },
})
```

2. **Configure degraded mode**:

```
verifier := agentauth.NewVerifier(&agentauth.VerifierConfig{
    DegradedMode: &agentauth.DegradedModeConfig{
        Enabled: true,
        OnRevocationServiceDown: agentauth.AllowCached,
        MaxCacheAge: 15 * time.Minute,
    },
})
```

17.2 10.2 Frequently Asked Questions

17.2.1 Q: Can AgentAuth work without a central server?

A: Yes, with limitations.

AgentAuth supports **offline verification** (Degraded Mode): - PoA tokens are self-contained - Signature verification works offline - Constraint checking works offline - Revocation checks require cache or online service

For truly air-gapped systems: 1. Pre-populate revocation cache

2. Use short-lived PoAs (e.g., 1-hour validity) 3. Accept risk of delayed revocation

17.2.2 Q: How does AgentAuth handle agent key rotation?

A: Built-in key rotation protocol.

```
// Rotate agent key

oldKeyID := "agent-key-2024"

newKeyID := "agent-key-2025"



// 1. Generate new key

newKey, _ := agentauth.GenerateEd25519Key()
```

```
// 2. Update entity profile

profile.PublicKeys = append(profile.PublicKeys, agentauth.PublicKey{

    KeyID: newKeyID,

    Algorithm: "Ed25519",

    KeyBytes: newKey.Public(),

    ValidFrom: time.Now().Add(7 * 24 * time.Hour), // 1 week overlap

})



// 3. Sign update with old key

signedUpdate, _ := profile.Sign(oldKey)
```

```
// 4. Publish update

entityService.Update(signedUpdate)



// 5. After overlap period, revoke old key

entityService.RevokeKey(oldKeyID)
```

17.2.3 Q: What happens if the principal's key is compromised?

A: Emergency revocation protocol.

1. **Immediate**: Report compromise to revocation authority
2. **Fast (5 min)**: All PoAs issued by compromised key are revoked

3. **Cascade**: All delegated PoAs also revoked

4. **Recovery (1 day)**: New key issued, delegations recreated

Prevention: - Use HSM for principal keys - Enable multi-signature for high-value principals - Regular key rotation

17.2.4 Q: Can PoA tokens be transferred?

A: No, by design.

PoA tokens are: - Bound to specific agent identity - Bound to specific principal - Non-transferable

Attempting transfer results in signature verification failure.

Exception: Delegation to sub-agents (requires principal con-

sent).

Chapter 18

Chapter 11: Future Directions

18.1 11.1 Post-Quantum Cryptography

AgentAuth is preparing for the post-quantum era.

18.1.1 Current Status

Vulnerable algorithms: - Ed25519 (broken by Shor's algorithm) - ECDSA P-256/384 (broken by Shor's algorithm)

Quantum-resistant algorithms: - Dilithium-3 (lattice-based) - SPHINCS+ (hash-based)

18.1.2 Roadmap

Phase 1 (2026): Hybrid signatures

```
{
  "signature": {
    "algorithm": "hybrid",
    "classical": {
```

```
      "algorithm": "Ed25519",
      "value": "..."
    },
    "post_quantum": {
      "algorithm": "Dilithium-3",
      "value": "..."
    }
  }
}
```

Both signatures must verify for token to be valid.

Phase 2 (2027): Pure post-quantum

```
{
  "signature": {
    "algorithm": "Dilithium-3",
    "value": "..."
  }
}
```

18.1.3 Implementation Guide

```
// Enable hybrid signing
signer := agentauth.NewSigner(&agentauth.SignerConfig{
    Mode: agentauth.HybridMode,
    ClassicalKey: ed25519Key,
```

```
    PostQuantumKey: dilithiumKey,
})

poa, _ := signer.Sign(poaData)

// Verifier automatically handles hybrid
verifier := agentauth.NewVerifier(&agentauth.VerifierConfig{
    SupportedAlgorithms: []string{"Ed25519", "Dilithium-3", "Hybrid"},
})

result, _ := verifier.Verify(ctx, poa)
```

18.2 11.2 Zero-Knowledge Proofs

Use case: Prove constraints without revealing values.

Example: Prove "transaction < limit" without revealing exact amount.

```
// Generate ZK proof
proof := agentauth.GenerateZKProof(&agentauth.ZKProofRequest{
    Statement: "transaction_amount < liability_cap",
    PrivateInputs: map[string]interface{}{
        "transaction_amount": 45000,
    },
    PublicInputs: map[string]interface{}{
```

```
        "liability_cap": 50000,

    },

})
```

```
// Verify ZK proof

valid := agentauth.VerifyZKProof(proof)
```

Status: Research phase, targeting 2027 release.

18.3 11.3 Multi-Agent Coordination

Vision: Agents negotiating with other agents.

Scenario: - Agent A wants to purchase from Agent B - Both agents verify each other's authority - Transaction executes only if both PoAs are valid

```
// Multi-agent transaction

func NegotiateTransaction(agentA, agentB *agentauth.Agent) (*Transaction, er

    // Both agents present PoAs

    poaA := agentA.GetPoA()

    poaB := agentB.GetPoA()



    // Cross-verify

    validA, _ := agentB.VerifyCounterparty(poaA)

    validB, _ := agentA.VerifyCounterparty(poaB)
```

```
    if !validA || !validB {
        return nil, fmt.Errorf("mutual verification failed")
    }

    // Execute transaction
    tx := &Transaction{
        Buyer: agentA.GetPrincipal(),
        Seller: agentB.GetPrincipal(),
        Amount: negotiatedAmount,
        Signatures: []Signature{
            agentA.Sign(txHash),
```

```
            agentB.Sign(txHash),
        },
    }

    return tx, nil
}
```

Status: Experimental, available in AgentAuth v2.0 (Q2 2026).

Chapter 19

Conclusion: The Path Forward

We stand at an inflection point in computing history. For the first time, software can act autonomously on behalf of humans—

not just execute commands, but make decisions, sign contracts, move money.

This power demands accountability. OAuth gave us access control. AgentAuth gives us legal accountability.

The techniques in this book—delegation chains, cryptographic proofs, liability constraints—are not theoretical. They are production-ready, battle-tested, and compliant with German, US, and EU law.

The choice is ours:

We can continue deploying agents with access tokens, hoping nothing goes wrong.

Or we can deploy agents with legal mandates, knowing exactly who is responsible when things do go wrong.

The agent's signature is not just a cryptographic value. It is a bridge between silicon and law, between algorithm and accountability.

When we sign, we accept responsibility. When agents sign, civilization scales.

END OF MANUSCRIPT

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